From snapshots to spatiotemporal patterns: Analysing Dynamic Edge Participation in Preterm vs. Term Neonates

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Introduction

Preterm infants show an increased risk for later neurodevelopmental disorders, including cognitive impairments and psychiatric disorders, likely linked to alterations in brain development (Fang et al., 2025). Understanding how preterm birth affects early brain maturation is essential for identifying developmental vulnerabilities and enabling the design of early interventions.

Building on the previous study by Fang et al. (2025), which examined static edge participation coefficients (ePC) in preterm and full-term neonates, this work explores a dynamic extension of their approach. We investigate temporal fluctuations in network integration during resting-state fMRI (rs-fMRI). This dynamic perspective aims to provide a more fine-grained and comprehensive understanding of early functional brain development and its alterations associated with preterm birth.

Methods

Dataset: Neonatal rs-fMRI (4th Release, Developing Human Connectome Project) (Edwards et al., 2022)
Sample (n = 796): Full-term: 514, Preterm scanned at postmenstrual age (PMA) $\geq$ 37 weeks: 133 (Preterm 1), Preterm scanned at PMA $<$37 weeks: 149 (Preterm 2)

Analysis pipeline

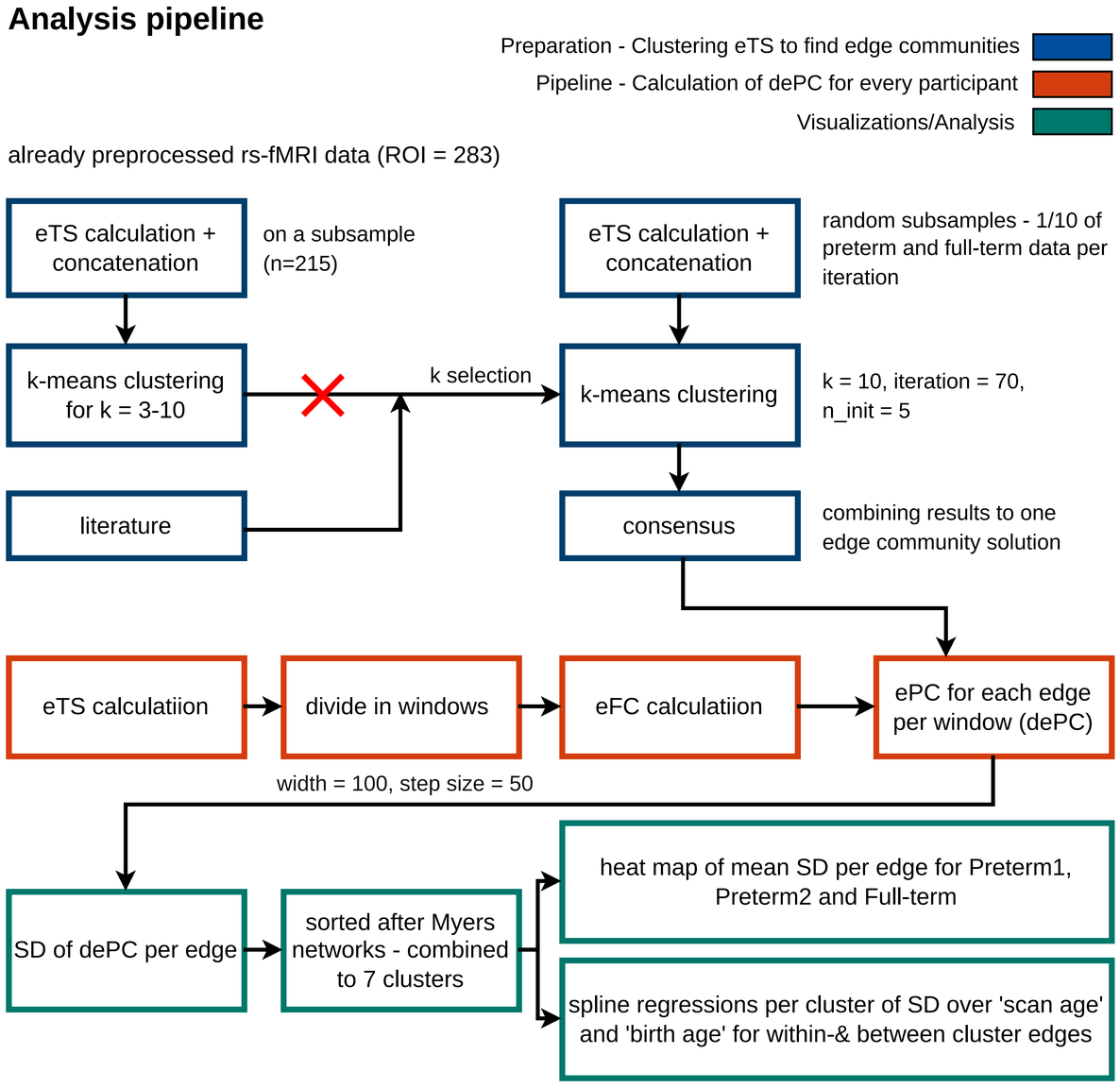

Figure 1: Dataset: Edwards et al. (2022), Atlas: Myers et al. (2024), SD = standard deviation, eTS = edge timeseries, eFC = edge functional connectivity

Dynamic Edge Participation Coefficient (dePC):
how strongly an edge shifts between segregated and integrated community roles over time.
Higher standard deviation (SD) $\rightarrow$ greater functional flexibility

Results

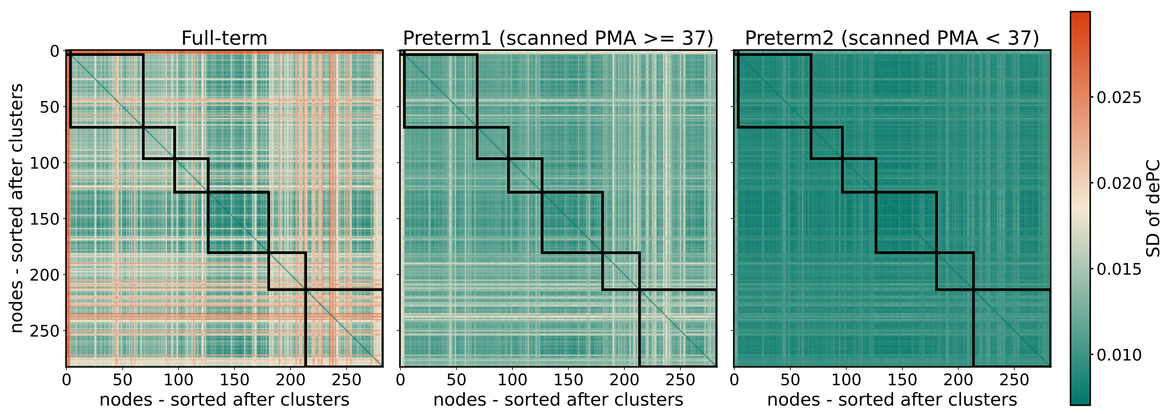

Figure 2: Full-term: 514, Preterm 1 - scanned at postmenstrual age (PMA) $\geq$ 37 weeks: 133, Preterm 2 - scanned at PMA$<$37): 149. Nodes are ordered according to our own aggregation of the networks defined by Myers et al. (2024), are outlined with black squares and numbered as in Table 1.

Cluster	Networks
0	Unassigned
1	Posterior default mode, Anterior default mode, Superior precuneus, Inferior precuneus, Anterior precuneus
2	Posterior fronto-parietal, Anterior fronto-parietal, Inferior fronto-parietal, Precentral fronto-parietal, Lateral orbito-frontal
3	Ventral attention, Dorsal attention
4	Posterior premotor, Anterior premotor, Auditory, MotorHand, MotorMouth
5	Medial orbito-frontal, Salience, Cingulate, Insula
6	Medial visual, Lateral visual

Table 1: Aggregation of the 24 networks from Myers et al. (2024) into seven clusters, numbered for reference in the following visualizations.

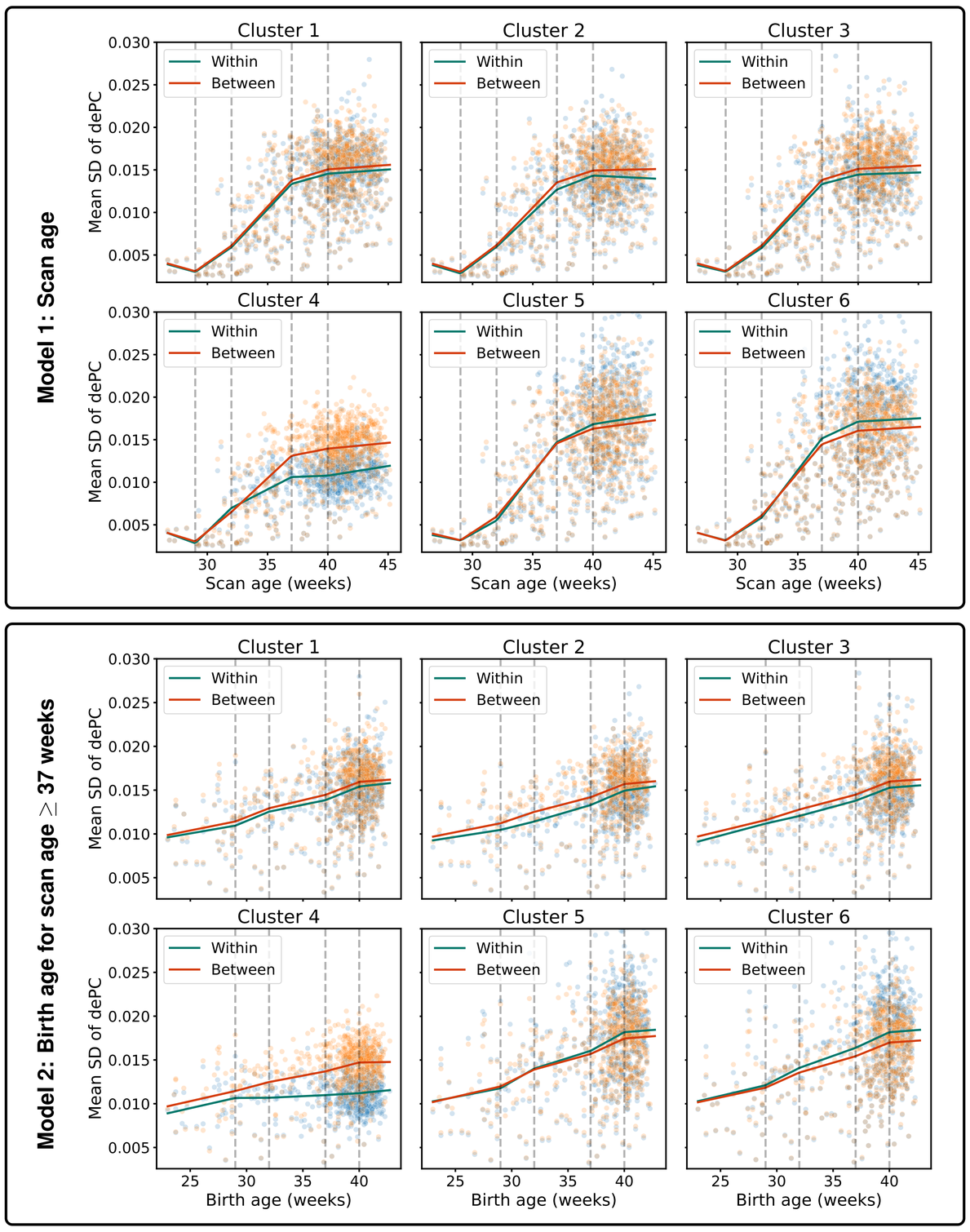

Figure 3: The figure shows the mean temporal SD of dePC for within- and between-network edges across participants, separated by network clusters 1–6. Within-network edges refer to connections among nodes belonging to the same functional network cluster, whereas between-network edges represent connections linking nodes from different network clusters. Model 1 displays trajectories as a function of scan age, while Model 2 shows trajectories by birth age, excluding preterms scanned at PMA$<$37. The splines were generated using cut-points based on defined preterm categories.

Discussion

Increasing functional flexibility: Term infants exhibit greater temporal variability in community integration compared to preterms.
Critical developmental window (29–37 weeks PMA): Period of rapid reorganization in functional communities, during which dynamic variability increases.
Maturation effects: Scan age (PMA at scan) is more predictive of dynamic community organization than birth age, although prematurity introduces a delay in maturation trajectories.
Sensorimotor network (SMN): Cluster 4, associated with the sensorimotor network shows lower functional flexibility compared to higher-order networks, but its dynamic integration into the global network increases with gestational age.

Limitations & Further Research

The static ePC effects reported by Fang et al. (2025) could not be replicated here, likely due to methodological differences; future work should test whether dePC is more robust to analytic choices and whether it improves prediction of later neurodevelopmental outcomes.

References

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